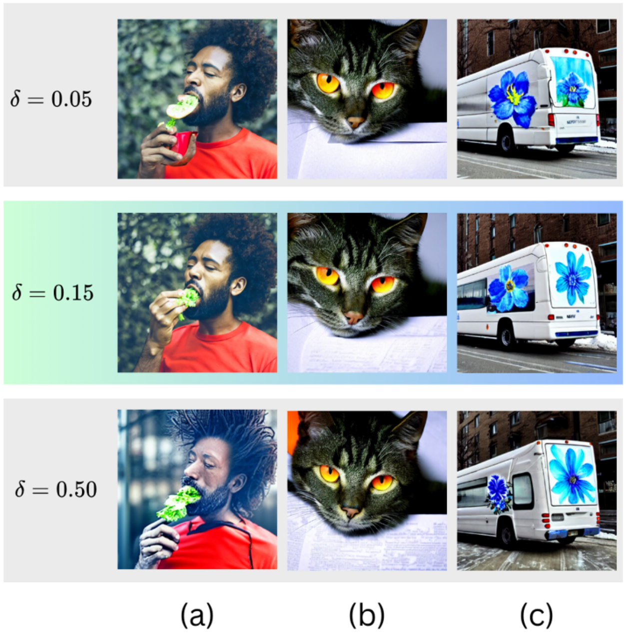

Figure 7. **Ablation demonstration for PAC Regularizer's** δ (a) a man in red shirt eating something that is green; (b) A green cat with orange eyes is laying over a paper; (c) A white bus with a painting of a blue flower on the front stopped on a street near a snow-covered sidewalk.

By incorporating this into the PAC-Bayes bound in Equation (13), we introduce the PAC-Bayes regularizer:

$$\mathcal{R}_{\text{PAC}} = -\sqrt{\frac{D_{\text{KL}}(A \parallel U) + \ln\left(\frac{2\sqrt{N}}{\delta}\right)}{2N}}. \tag{20}$$

Piecing everything together, our final loss function becomes:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{div}}\mathcal{L}_{\text{div}} + \lambda_{\text{sim}}\mathcal{L}_{\text{sim}} + \lambda_{\text{out}}\mathcal{L}_{\text{out}} + \lambda_{\text{PAC}}\mathcal{R}_{\text{PAC}}, \tag{21}$$

where $\lambda_{\text{div}} = \alpha$, $\lambda_{\text{sim}} = \beta$, $\lambda_{\text{out}} = \gamma$, and $\lambda_{\text{PAC}} = \eta$.

This formulation ensures that our loss function not only captures the empirical performance (through the divergence, similarity, and outside losses) but also incorporates theoretical guarantees on generalization provided by the PAC-Bayes framework.

D. Implementation Details

Our experiments were conducted on a system equipped with 4 NVIDIA A6000 GPUs capable of handling the computational demands of diffusion models and attention map computations. We utilized the official Stable Diffusion v1.4 model [35] and the CLIP ViT-L/14 text encoder [31] for text embeddings.

For hyperparameters, we used a step size $\alpha_t = 20$, and the denoising steps was set to $T = 50$. The loss weights were set as $\lambda_{\text{div}} = -1.25$, $\lambda_{\text{sim}} = 2.0$, $\lambda_{\text{out}} = 0.15$, and $\lambda_{\text{PAC}} = -0.15$.

We employed the spaCy library [17] with the `en_core_web_trf` transformer model to parse the text prompts. The parser identifies nouns and their associated modifiers based on syntactic dependencies, enabling us to construct the sets $\mathcal{M}_k$ and $\mathcal{N}_k$. Attention maps were extracted from the cross-attention layers during the denoising process and normalized to ensure they are valid probability distributions over the spatial dimensions.

We used gradient descent to update the latent variables z_t based on the computed total loss. Gradients were computed with respect to z_t, and updates were applied during the specified timesteps. To mitigate computational impact, we limited the latent updates to the first half of the denoising steps and optimized the implementation of attention map computations and loss evaluations.

E. Computational Efficiency

Method	SD	AnE	SG	EBAMA	Ours
Time (min)	13.2	47.0	21.9	20.9	35.7

Our method introduces additional computations due to the extraction and manipulation of attention maps and the application of custom loss functions during the denoising steps. As a result, the generation time per image increases compared to baseline models like Stable Diffusion. Despite this overhead, our method remains computationally feasible and can be optimized further; the trade-off in speed is justified by the significant improvements in image quality and attribute-object alignment.

F. Additional Ablation Experiments

In this section, we analyze the impact of varying key hyperparameters: the PAC-Bayes confidence parameter δ, the number of updated timesteps T', and the step size α. Visual comparisons are provided in Figures 7, 8, and 9, respectively, using the prompts from ABC-6K dataset.

Confidence parameter δ We evaluate δ values of 0.05, 0.15 (ours), and 0.50 as shown in Figure 7. At $\delta = 0.05$, strong regularization leads to underfitting, resulting in poorly rendered objects: the green object in (a), the unnatural green cat in (b), and the faint blue flower in (c). Conversely, $\delta = 0.50$ weakens regularization, causing overfitting with dominant attributes overshadowing others: the red shirt in (a), excessive orange eyes in (b), and overly prominent blue flower in (c). Our choice of $\delta = 0.15$ balances regularization and flexibility, ensuring accurate attribute-object alignment and visual coherence.